Intro to Online Optimization and Learning

CS245: Online Optimization and Learning

Xin Liu
SIST, ShanghaiTech University

General information about this course

General information:

- 12 weeks and 3 credits
- "Prerequisites": Algorithms and data structures, Probability and statistics, and Convex optimization

Grades:

- Three problem sets (30%) : algorithm implementation (with python) and analysis (with latex).
- Final project (60%) : report (45%) and presentation (15%). Group (≤ 3) and indicate your individual contribution.
- Participation and quiz: 10%.

Late homework policy:

- 3 late days ($75\% \to 50\% \to 25\%$) for homework and no late days for the final project!

General information about this course

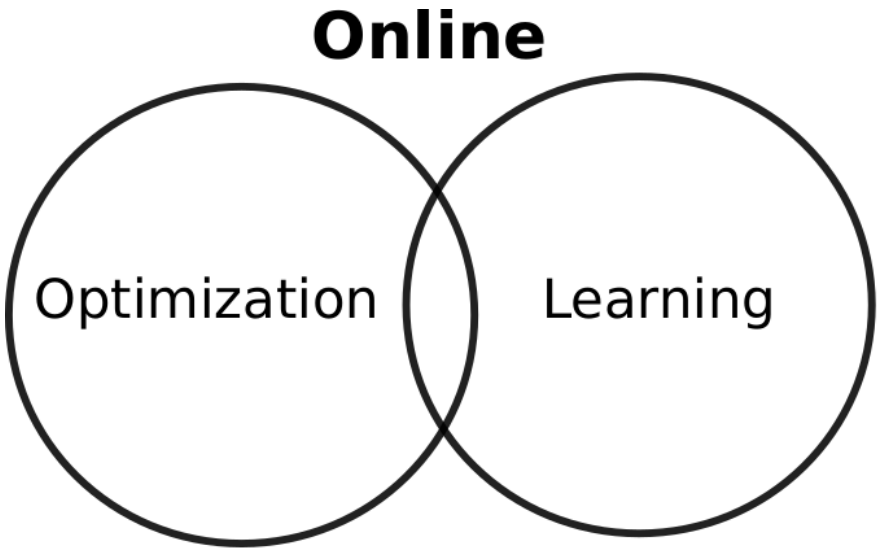
TA: Hengquan Guo (guohq@shanghaitech.edu.cn), 2-209, SIST

Office hour: 10:00–11:00am on Friday at 2-302H, SIST

Academic Integrity!!!

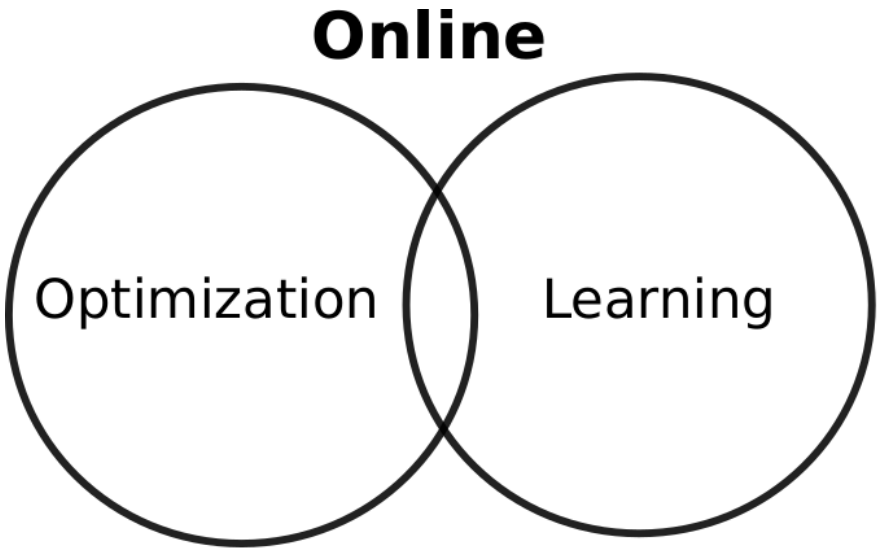
- ShanghaiTech academic integrity policy.
- Problem sets: be indepedent, feel free to discuss, and cite when you borrow ideas from books, internet, and peers, etc.
- Final project: indicate your individual contribution in the report.

Course Overview



What does "online" mean?

Course Overview



What does "online" mean?

- Antonym of offline/batch

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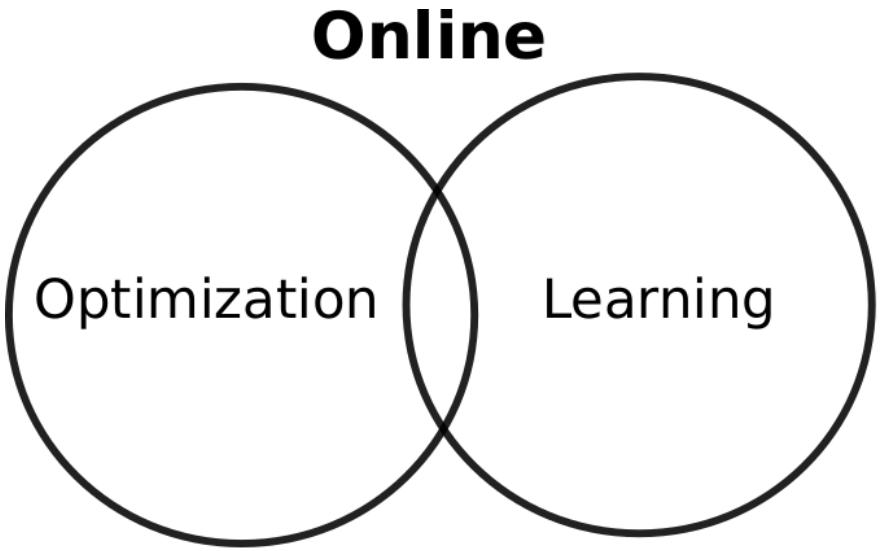

What does "online" mean?

- Antonym of offline/batch
- Stochastic and uncertainty

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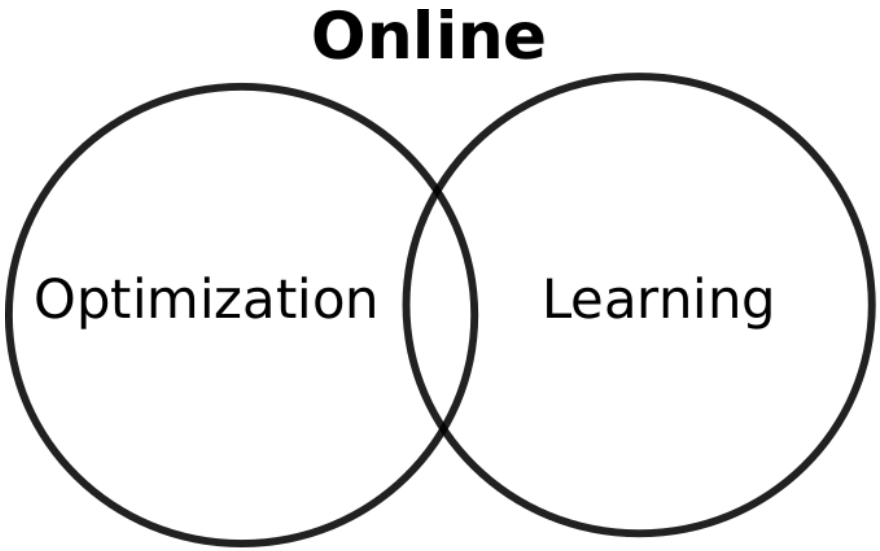

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- Antonym of offline/batch
- Stochastic and uncertainty
- Adaptive, efficient, and robust

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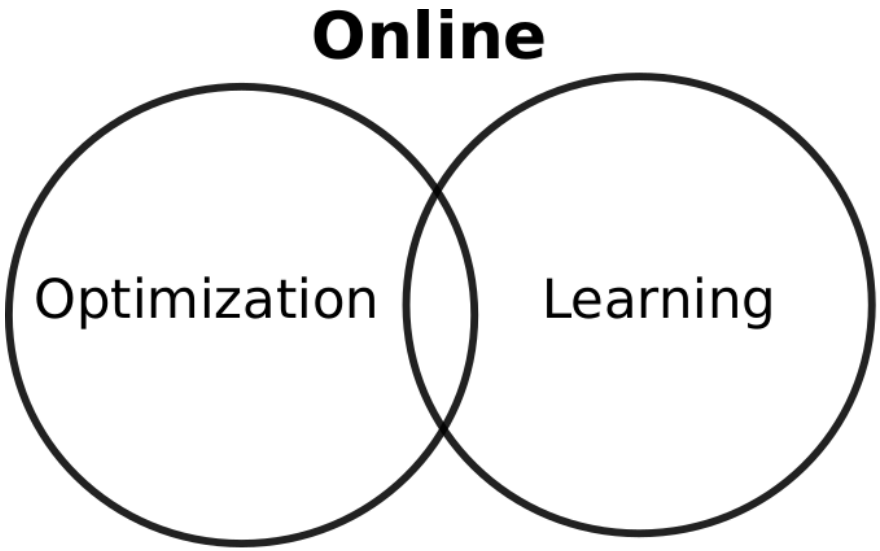

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- Stochastic and uncertainty
- Adaptive, efficient, and robust
- Challenging and powerful

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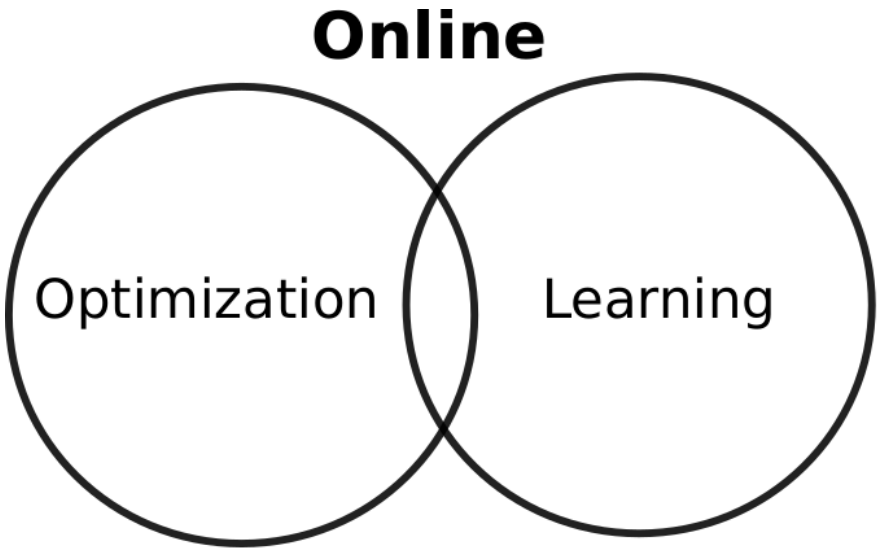

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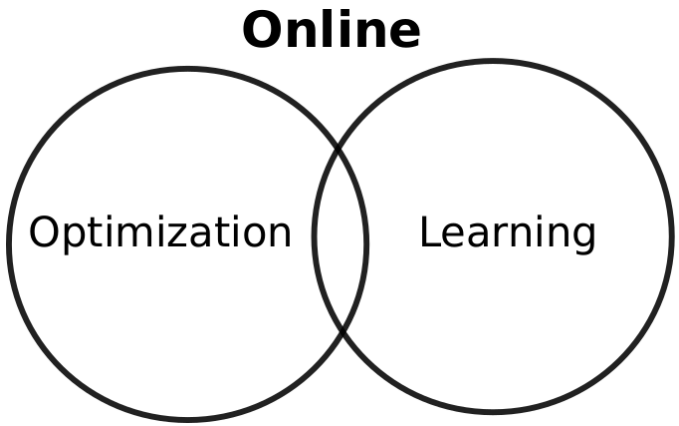
Online decision-making under uncertainty

Course Overview



Applications

- Recommendation systems
- AI gaming (Alpha Go)
- Healthcare (clinic trial)
- Wireless nets (resource allocation)
- LLMs (finetuning/reasoning)
- $\cdots$ Your Research $\cdots$

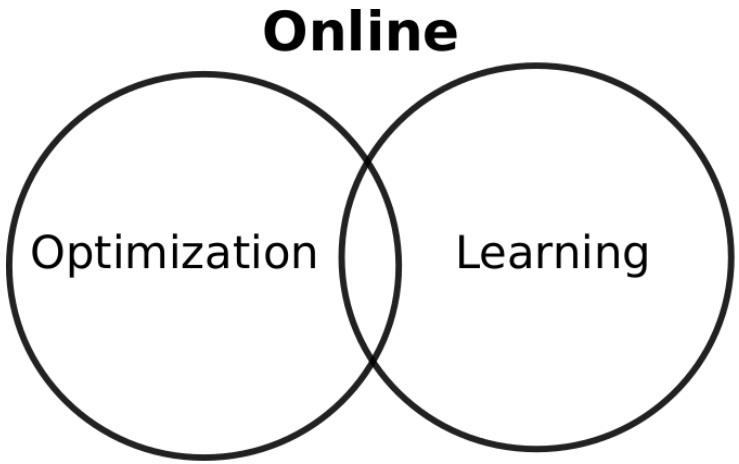
Theory

- Online convex optimization
- Stochastic/Adversival Bandits
- Reinforcement learning

Algorithms

- Online gradient descent
- Upper confidence bound

Course Overview



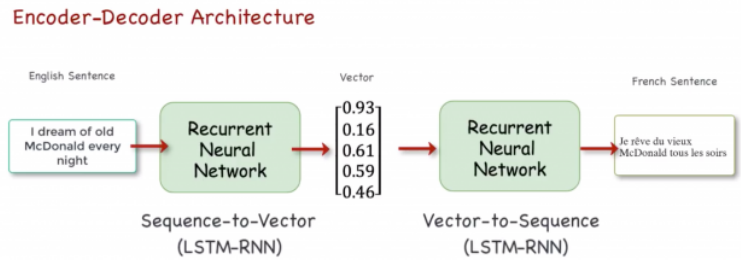
The style of the courses:

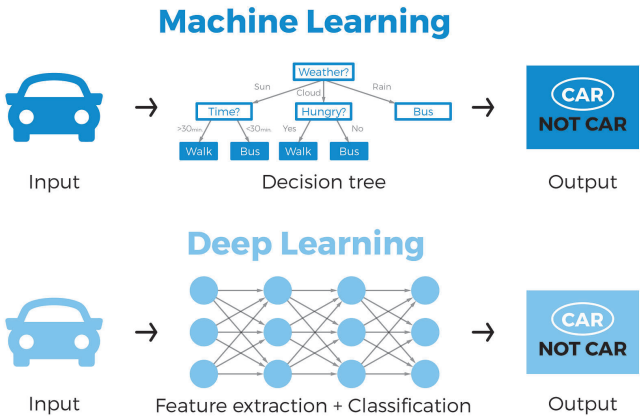
- Motivated examples
- Problem formulation via online O&L
- Intuition and online algorithms
- Performance analysis (e.g., regret)
- Discussions (e.g., optimality)

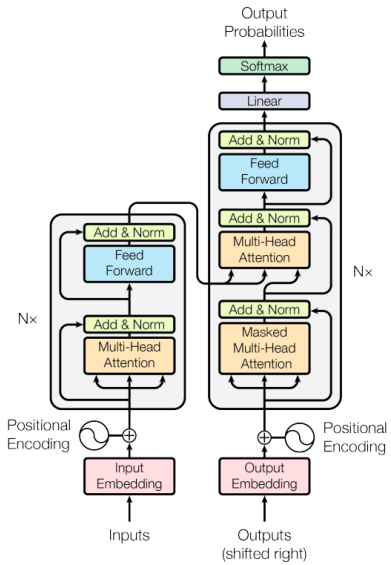
Offline learning

We know (offline or batch) learning:

- Objective recognition
- Machine translation







Key elements in (offline or batch) learning:

- Data
- Models and algorithms

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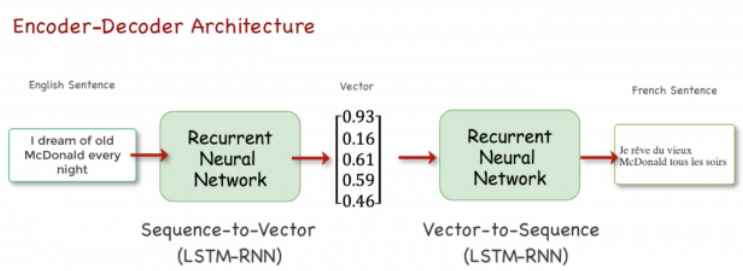

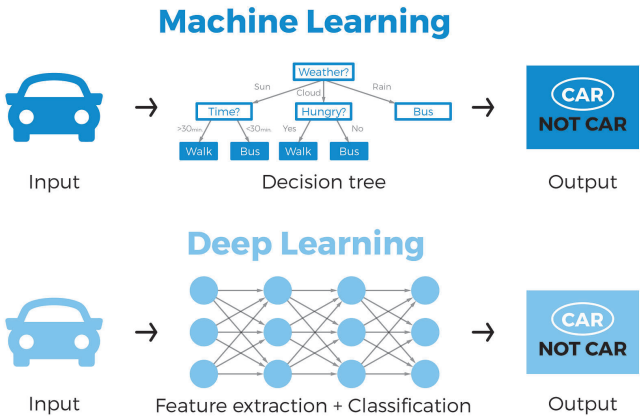

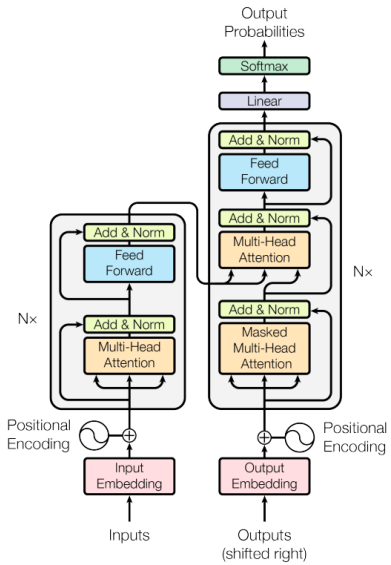

(Offline or batch) learning:

- Data is important, which is given and fixed (ImageNet, SQuAD).
- Tune models or algorithms (linear or non-linear, structure of deep nets).

Offline Optimization

We know (offline) optimization:

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & g(x) \leq 0 \end{aligned}$$

How to solve an optimization problem?

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- We solve it with solvers (e.g., CVX, Gurobi)

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- We solve it with solvers (e.g., CVX, Gurobi)
- We like "linear" and run simplex algorithm
- We like "convex" and check KKT condition

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How to solve an optimization problem?

- We solve it with solvers (e.g., CVX, Gurobi)
- We like "linear" and run simplex algorithm
- We like "convex" and check KKT condition
- Just run (stochastic) gradient descent
- .....

A well-known example of offline learning and optimization

Consider linear regression (LR) for:

- Shanghai Putong house price prediction

Given historical/batch data $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \cdots (\mathbf{x}_T, y_T)\}$, we do LR

$$\min_{\mathbf{w}} \|\mathbf{X}^T\mathbf{w} - \mathbf{y}\|_2^2$$

Solve LR:

- in a close form?
- with (stochastic) gradient descent
- with regularizers always

Goal:

avoid overfitting

&

min the generalization error.

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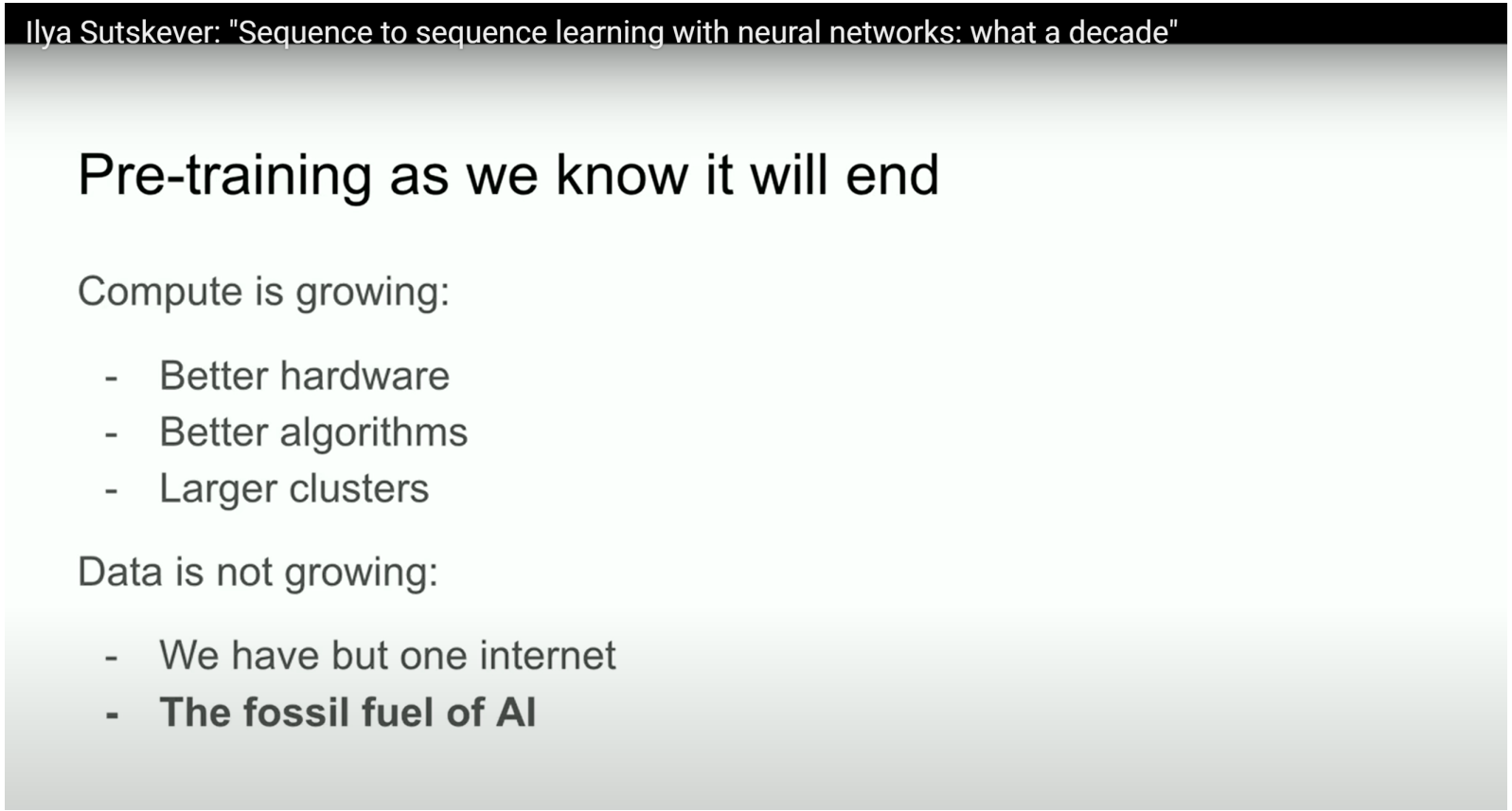
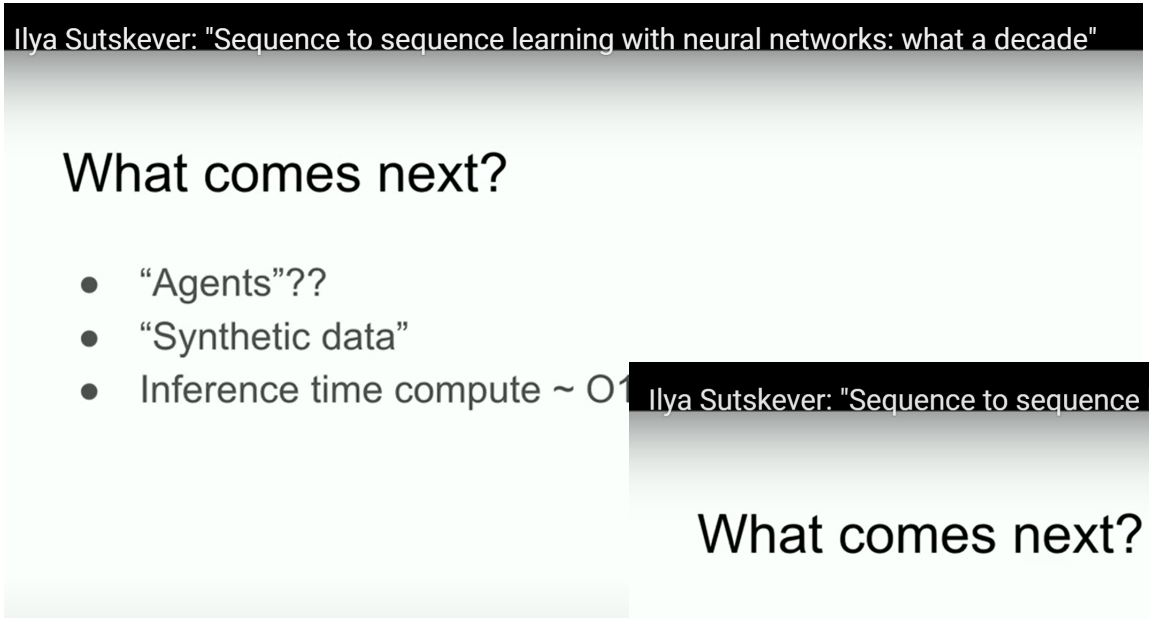
How about we gain data in an online way (with distribution drift)?
How about your model (policy) affects your data?

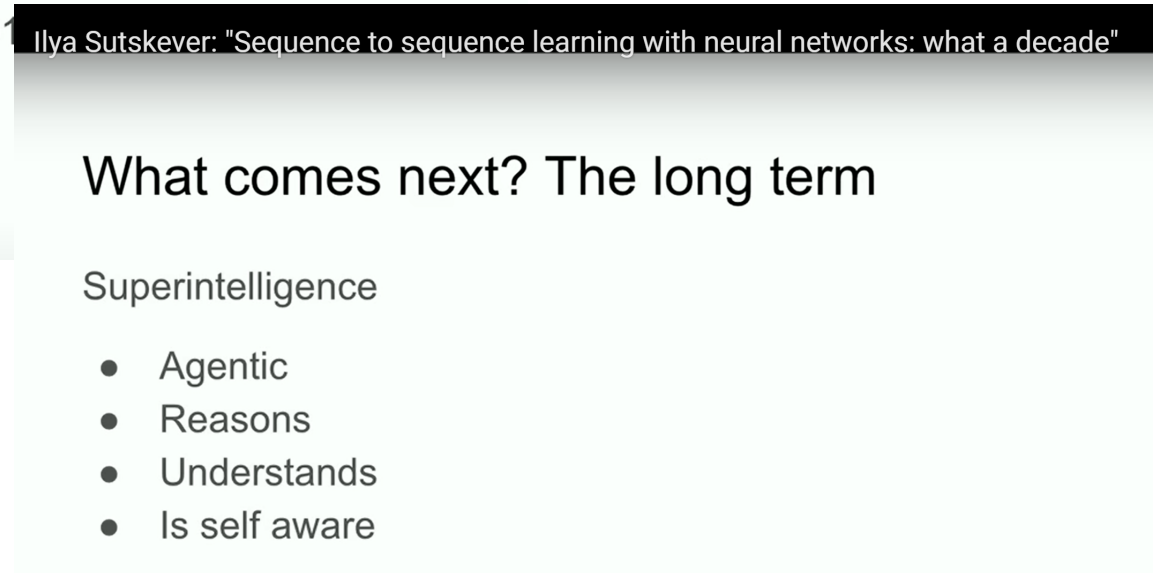
Offline learning is ending?



How about we do not have any more data?
"Offline training or pretraining is ending?!" Illya at NeurIPS24.

From offline to online learning?



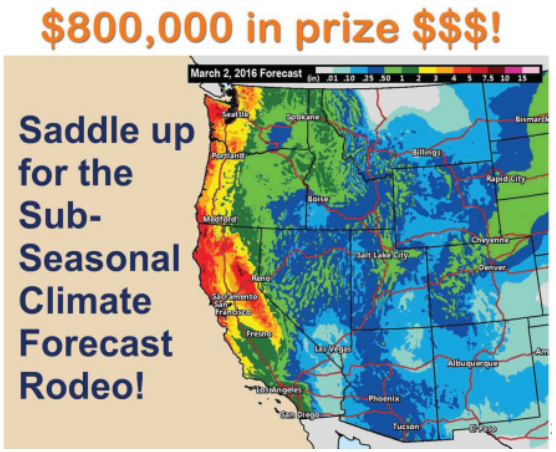


What comes next? online, continual, and reinforcement learning?

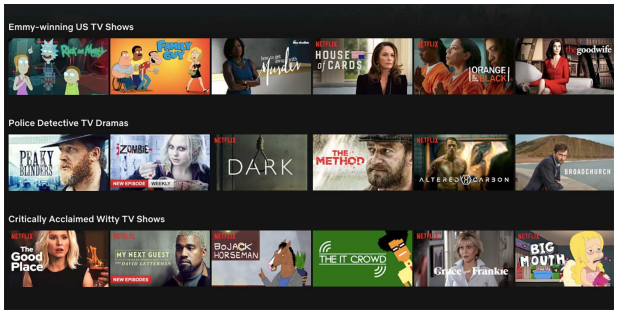
Motivated examples

Online optimization and learning:

- Movie recommendation
- Go game
- Climate prediction
- LLMs



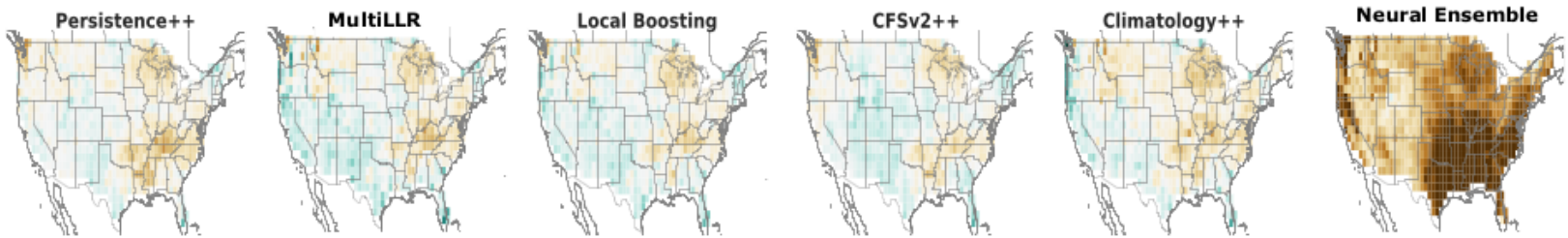
deepseek



Subseasonal climate forecast[1]

Subseasonal Climate Forecast Rodeo by the Bureau of Reclamation and the National Oceanic and Atmospheric Administration:

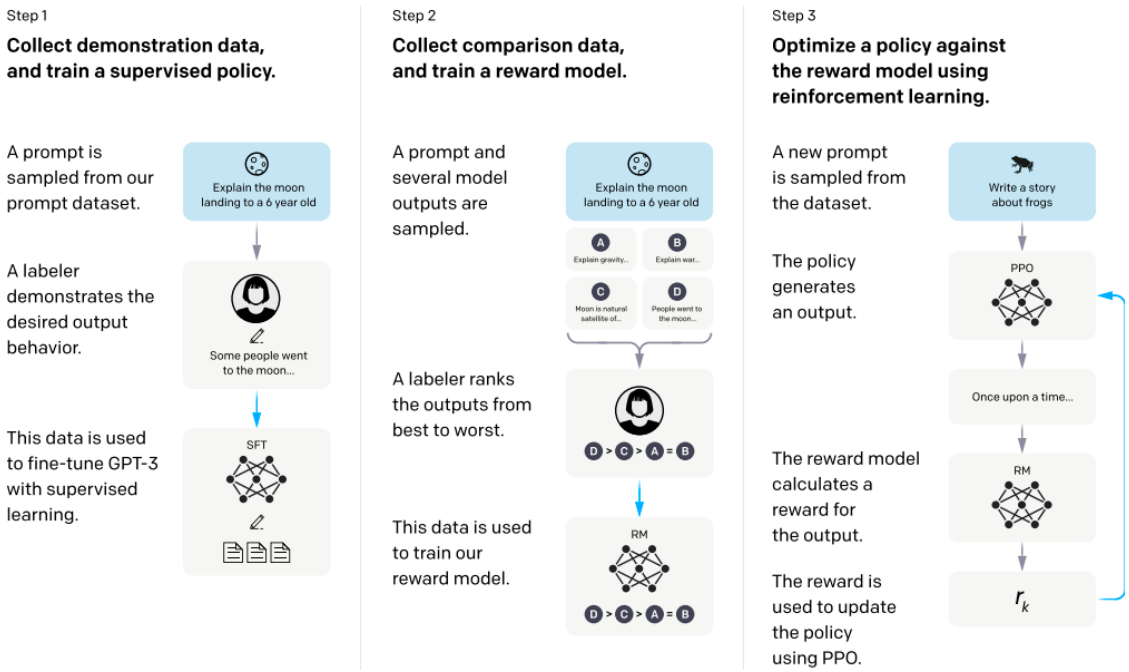
- Temperature and precipitation of $2 \sim 6$ weeks in the Western US (Holy grail of forecasting).
- Diverse collection of forecasting models.
- At least one model performs well each year (but unclear which apriori).



Pefectly fit online optimization and learning?

[1] Genevieve E Flaspohler, etc,. "Online Learning with Optimism and Delay". ICML, 2021.

LLM Finetuning



"Reinforcement learning from human feedback" for LLM finetuning[2]:

- Learn a base/ref policy from supervise learning
- Learn reward models via human/AI feedback
- Policy optimization based on reward models

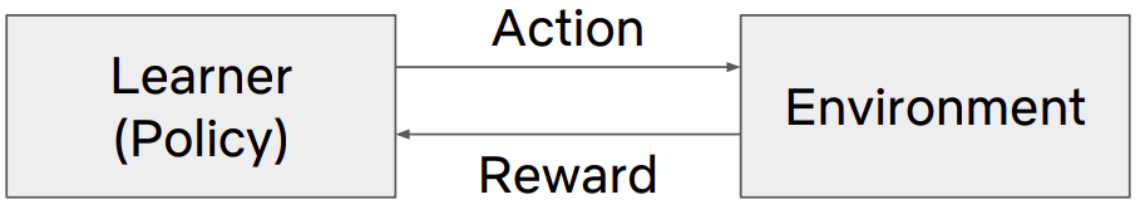
[2] Long Ouyang, etc,. "Training language models to follow instructions with human feedback". NeurIPS, 2022.

A first look at online learning

Online learning framework

For $t = 1, \cdots, T$:

- **Learner:** Submit action/decision x_t.
- **Environment:** Observe the feedback: loss $f_t(\cdot)$ or reward $r_t(\cdot)$.



A first look at online learning

Online learning framework

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For different applications, $x_t, f_t(\cdot), r_t(\cdot)$ have different meaning:

- subseason climate prediction
- movies/items recommendation
- matching in crowdsourcing
- resource allocation in wireless system
- LLMs finetuning/reasoning
- .....

A first look at online learning

Online learning framework

For $t = 1, \cdots, T$:

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The regret of a learner is defined to be:

$$\text{Regret}(T) = \sum_{t=1}^{T} f_t(x_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^{T} f_t(x),$$

$$\text{Regret}(T) = \max_{x \in \mathcal{X}} \sum_{t=1}^{T} r_t(x) - \sum_{t=1}^{T} r_t(x_t).$$

Objective of learner: minimize the Regret(T)!

A first look at online learning: subseasonal climate forecast

Aggregating weather prediction:

Initialization: N experts/models.
For each time period $t = 1, \cdots, T$:

- **Learner:** Combine predictions from N experts/models and make a prediction on the weather.
- **Environment:** Observe the outcome.

Objective: find the best expert in hindsight and minimize the regret as follows

$$\text{Regret}(T) = \sum_{t=1}^{T} f_t(x_t) - \sum_{t=1}^{T} f_t(\text{best expert})$$

Prediction with Expert Advice

Aggregating weather prediction:

Initialization: N experts/models.

For each day $t = 1, \cdots, T$:

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Objective: follow the best expert advice (without knowing the best expert)

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- How to combine the advices from N experts?
- How to update the belief on these experts? Exclude bad experts and find good experts?

Prediction with Expert Advice

Prediction with weighted majority algorithm (WMA):

Initialization: $w_1(i) = 1, \forall i \in [N]$.

For each day $t = 1, \cdots, T$:

- **Learner:** if $\sum_{i \in S_t(A)} w_t(i) \geq \sum_{i \in S_t(B)} w_t(i)$ then choose A; otherwise choose B.
- **Environment:** Observe the outcome if it is A or B.
- **Update:** $w_{t+1}(i) = (1 - \epsilon) w_t(i)$ for these experts that make mistake.

Intuition: trust experts with large weights

Prediction with Expert Advice

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Intuition: trust experts with large weights

- Combine the advices by "majority voting".
- Update the belief with multiplicate/exponenital decay with learning rate ϵ.

Prediction with Expert Advice

Define M_T and $M_T(i)$ to be the total # of mistakes the WMA and the expert i make until T, respectively.

Theorem 1

For any expert $i \in [N]$, the weighted majority algorithm with the learning rate $(0 < \epsilon < 0.5)$ achieves

$$M_T < 2(1+\epsilon)M_T(i) + \frac{2\log N}{\epsilon}.$$

- # of mistakes the weighted majority algorithm is approximately twice # of mistakes the best expert makes.

According to Alg update:

$w_{T+1}(i)$ encodes the info of $M_T(i)$

$\sum_i w_{T+1}(i)$ encodes the info of M_T

Prediction with Expert Advice – Proof

A "potential/Lyapunov drift" style of analysis: define

$$\phi_t = \sum_{i \in [N]} w_t(i),$$

and study the drift

$$\phi_{t+1} - \phi_t \text{ or } \phi_{t+1}/\phi_t.$$

Two observation: $\phi_1 = N$

$\phi_{t+1} \leq \phi_t$ (a key property)

Recall we want to find the relation between m_T and $m_T(i)$,

which can be connected by ϕ_T.

Prediction with Expert Advice – Proof

We only consider the case when WMA makes a mistake:

1) WMA predicts A but B returns

2) WMA predicts B but A returns

Let's consider case 1) and please verify 2) by yourself.

Since WMA choose A, we have

$$\sum_{i \in S_t(A)} w_t(A) > \frac{\phi_t}{2} \qquad \text{eq-1}$$

We study the drift:

$$\phi_{t+1} = \sum_i w_{t+1}(i)$$

$$= \sum_{i \in S_t(A)} w_{t+1}(i) + \sum_{i \in S_t(B)} w_{t+1}(i)$$

$$= (1-\epsilon) \sum_{i \in S_t(A)} w_t(i) + \sum_{i \in S_t(B)} w_t(i)$$

$$= \phi_t - \epsilon \sum_{i \in S_t(A)} w_t(i) \qquad \text{[by eq-1]}$$

$$\leq \phi_t \left(1 - \frac{\epsilon}{2}\right)$$

Note the inequality holds only if the Algorithm made a mistake at time t, otherwise we have $\phi_{t+1} \leq \phi_t$.

Recall the definition of m_T, we have

$$\phi_{T+1} \leq \left(1 - \frac{\epsilon}{2}\right)^{m_T} \cdot \phi_1$$

$$= \left(1 - \frac{\epsilon}{2}\right)^{m_T} \cdot N$$

Moreover, we have

$$\phi_{T+1} \geq w_{T+1}(i)$$

$$= (1-\epsilon)^{m_T(i)} \cdot w_1(i)$$

$$= (1-\epsilon)^{m_T(i)}$$

Finally, we have

$$(1-\epsilon)^{m_T(i)} \leq \left(1 - \frac{\epsilon}{2}\right)^{m_T} \cdot N$$

Take "log" on both sides:

$$m_{T(i)} \log(1-\epsilon) \leq m_T \log\left(1-\frac{\epsilon}{2}\right) + \log M \qquad \text{eq-2}$$

Now we need a key inequality that

$$-\epsilon - \epsilon^2 \leq \log(1-\epsilon) \leq -\epsilon, \quad \forall \epsilon \in (0, 0.5) \qquad \text{eq-3}$$

please verify it by yourself.

Therefore, combine eq-2 and eq-3, we have

$$m_{T(i)}(-\epsilon - \epsilon^2) \leq m_T\left(-\frac{\epsilon}{2}\right) + \log M,$$

which completes the proof.

Expert problem

Expert problem:

Initialization: N experts/models.
For each day $t = 1, \cdots, T$:

- **Learner:** Obtain predictions from N experts/models and choose an expert with probability p_t.
- **Environment:** Observe the loss of each model $\ell_t \in [0,1]^N$.

Objective: Find the best expert in hindsight, which is equivalent to minimize regret:

$$\mathcal{R}(T) := \sum_{t=1}^{T} \langle p_t, \ell_t \rangle - \sum_{t=1}^{T} \langle p, \ell_t \rangle = \sum_{t=1}^{T} \langle p_t, \ell_t \rangle - \sum_{t=1}^{T} \ell_t(i^*)$$

Expert problem: Hedge

Hedge (Exponentially Weighted Average)

Initialization: $w_1(i) = 1, \forall i \in [N]$.
For each day $t = 1, \cdots, T$:

- **Learner:** Compute the prob of choosing expert i : $p_t(i) = w_t(i) / \sum_i w_t(i)$.
- **Environment:** Observe the loss of each model $\ell_t \in [0, 1]^N$.
- **Loss update:** $w_{t+1} = w_t \cdot e^{-\eta \ell_t(i)}, \forall i \in [N]$.

- p_t is just the softmax function of accumulative loss.
- A better expert with small accumulated loss $\sum_{s=1}^{t} \ell_s(i)$ will be selected with a high probability.
- Can we find the best expert with Hedge? What is $\mathcal{R}(T)$?

Expert problem: Regret of Hedge

Theorem 2

The regret of Hedge is

$$\begin{aligned}\mathcal{R}(T) \leq & \frac{\log N}{\eta} + \eta \sum_{t=1}^{T} \sum_{n=1}^{N} p_t(i) \ell_t^2(i) \\ \leq & \frac{\log N}{\eta} + T\eta.\end{aligned}$$

- Learning rate is important and let $\eta = O(\sqrt{\log N / T})$ provides $\mathcal{R}(T) = O(\sqrt{T \log N})$.
- $\lim_{T \to \infty} \mathcal{R}(T)/T = O(1/\sqrt{T}) \to 0$. Hedge can achieve the best expert asymptotically.

Expert problem: Regret of Hedge – proof

We study a "potential drift" proof with the potential function

$$\Phi_t = \sum_{i \in [N]} w_t(i).$$

The proof is similar with that of weighted majority algorithm.

We study ϕ_{t+1} as follows:

$$\phi_{t+1} = \sum_i w_t(i)\, e^{-\eta \ell_t(i)}$$

$$= \phi_t \sum_i \frac{w_t(i)}{\phi_t} e^{-\eta \ell_t(i)}$$

$$= \phi_t \sum_i p_t(i)\, e^{-\eta \ell_t(i)}$$

Expert problem: Regret of Hedge – proof

Note we want the loss of Alg $\sum_i p_i \ell(i)$.

Suppose we have

$$\sum_i p_t(i)\, e^{-\eta \ell(t,i)} \le e^{-\eta \sum_i p_t(i)\, \ell(t,i)} \qquad \text{eq-1}$$

like $E[e^{-\eta X}] \le e^{-\eta E[X]}$, where $X \sim \{\ell(t,i)\}$ according to prob $\{p_i\}$.

We are almost done because we already have

$$\phi_{t+1} \le \phi_t\, e^{-\eta \sum_i p_t(i)\, \ell(t,i)}$$

which implies

$$\phi_{T+1} \leq \phi_1 \, e^{-\eta \sum_t \langle p_t, \ell_t \rangle}$$

Moreover, we have

$$\phi_{T+1} \geq e^{-\eta \sum_t \ell_t(i)} \qquad \forall i$$

Therefore, we have

$$e^{-\eta \sum_t \ell_t(i)} \leq N \, e^{-\eta \sum_t \langle p_t, \ell_t \rangle}$$

Take "log" on both sides such that

$$R(T) = \sum_t \langle p_t, \ell_t \rangle - \sum_t \ell_t(i) \leq \frac{\log N}{\eta}$$

It is even better than thm 2

the regret is "constant", too good to be true!

Can you see where and why it fails?

(see the next page)

It is on $\sum_i p_t(i) e^{-\eta \ell_t(i)} \overset{?}{\leq} e^{-\eta \sum_i p_t(i) \ell_t(i)}$ eq-1

It is actually the Jensen's inequality, we have

$$\sum_i p_t(i) e^{-\eta \ell_t(i)} \geq e^{-\eta \sum_i p_t(i) \ell_t(i)} \quad !$$

Now we need some approximation as follows.

$$\phi_{t+1} = \phi_t \sum_i p_t(i) e^{-\eta \ell_t(i)}$$

$$\leq \phi_t \sum_i p_t(i) \left(1 - \eta \ell_t(i) + \eta^2 \ell_t^2(i)\right) \quad [e^{-x} \leq 1 - x + x^2, \ x > 0]$$

$$= \phi_t \left(1 - \eta \sum_i p_t(i) \ell_t(i) + \eta \sum_i p_t(i) \ell_t^2(i)\right)$$

$$\leq \phi_t \, e^{-\eta \sum_i p_t(i) \ell_t(i) + \eta \sum_i p_t(i) \ell_t^2(i)} \quad [1 + x \leq e^x]$$

Now we have the iteration on

ϕ_{t+1} and ϕ_t

please finish the proof by yourself!

Expert problem: Lower bound

We have derived the regret of $O(\sqrt{T \log N})$ for Hedge.

- Can we design even better policies/algorithms with the smaller regret?
- Is this improvable or optimal?

Expert problem: Lower bound

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- Can we design even better policies/algorithms with the smaller regret?
- Is this improvable or optimal?

To answer these questions, we need to study the lower bound.

- The lower bound indicates how hard the problem is.
- The lower bound is more about the problem itself and it is universal for any algorithms.
- Your policies/algorithms are good (optimal) if their performance are close to the lower bound.
- For Expert problem, what is the lower bound of regret? Is it $\Omega(\sqrt{T \log N})$?

Expert problem: Lower bound

Theorem 3

For any algorithm, we have

$$\lim_{T,N\to\infty} \max_{\ell_1,\ell_2,\cdots,\ell_T} \frac{\mathcal{R}(T)}{\sqrt{T\log N}} \geq \frac{1}{\sqrt{2}}.$$

- The asymptotically lower bound of expert problem is $O(\sqrt{T\log N})$.
- Hedge is asymptotically optimal!
- The proof can be found in that of Theorem 3.7 in the book "Prediction, Learning, and Games"[3].

[3] A more accessible proof is Theorem 3 of Lecture 1 in a related course "Introduction to Online Learning" https://haipeng-luo.net/courses/CSCI659.